Distributed Active Disturbance Rejection Formation Tracking Control for Quadrotor UAVs

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***Abstract*—In this article, a distributed active disturbance rejection formation tracking control strategy is developed for a quadrotor unmanned aerial vehicle (UAV) swarm system with switching communication topologies. The proposed control strategy consists of two parts: 1) attitude-loop control and 2) position-loop control. First, a distributed cascade active disturbance rejection control (CADRC) method is designed for the attitude subsystem. With this attitude control method, the stability of the attitude subsystem can be achieved even in the presence of unknown time-varying disturbances. Second, a distributed formation tracking control method is designed for the position subsystem. This position control method ensures that the quadrotor UAV swarm maintains dynamic formation flying and accurately tracks a predetermined trajectory. Through stability analysis, it can be proved that the proposed control strategy can ensure the stability of the whole swarm system. Finally, the proposed control strategy is applied to a quadrotor UAV swarm system to verify its effectiveness and the ability to suppress the influence of unknown time-varying disturbances.**

***Index Terms*—Cascade active disturbance rejection control (CADRC), distributed control, formation tracking control, unmanned aerial vehicle (UAV).**

I. Introduction

In RECENT years, quadrotor unmanned aerial vehicles (UAVs) have been widely used in military and civilian fields, such as agricultural mapping [1], wireless communication [2], and nuclear radiation detection [3], due to their advantages of mobility and flexibility, low cost, and simple structure. Researchers have developed many useful flight control methods for quadrotor UAVs, including PID control [4], adaptive backstepping control [5], and sliding-mode control [6]. In practical applications, a formation composed of multiple low-cost quadrotor UAVs can replace an expensive multifunctional UAV to complete complex tasks. Moreover, UAV formations can provide system redundancy and reconfiguration capabilities compared with an individual UAV [7]. In the past decade, quadrotor UAV formations have been applied in many practical scenarios, such as agricultural spraying [8] and forest firefighting [9].

According to the sensing capability of quadrotor UAVs, the existing quadrotor formation control results [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20] can be roughly divided into three groups: 1) position-based method [18]; 2) displacement-based method [19]; and 3) distance-based method [20]. Every method possesses distinct advantages with respect to sensing capability and interaction topology requirements for quadrotor UAVs. Selection among these methods is dependent on the specific circumstances of practical applications. In addition, from the perspective of control mechanisms, other effective quadrotor formation control methods have been developed, including the leader–follower method [21], virtual structure-based method [22], and behavior-based method [23]. These methods can be treated as special cases of consensus-based formation control methods [24]. In recent years, many consensus-based formation control methods for quadrotor UAVs have been proposed. For example, [25], [26], [27], [28], and [29] employed consensus-based approaches to address the time-varying formation tracking control challenges of quadrotor UAVs. However, these approaches rely on a simplified model of the quadrotor. In [30], a consensus-based cooperative formation control scheme with collision avoidance was developed for a multi-UAV system. In [31], a super twisting algorithm was designed to achieve robust consensus-based formation flight of multiple quadrotors. Note that the design of these formation control methods relies on the velocity information of quadrotor UAVs. For certain small and low-cost quadrotor UAVs, velocity sensors might not be installed due to tight space and cost constraints. As a result, designing a formation control method that relies solely on position information for quadrotor UAVs becomes significant. This is the first motivation of this article.

In general, quadrotor UAVs are highly sensitive to unknown disturbances due to their small size and lightweight. As a result, a variety of advanced disturbance rejection control schemes have been developed for quadrotor UAVs [32], [33],

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[34], [35], [36], [37]. For instance, Wang and Su [36] designed a robust nonlinear feedback control strategy for an aircraft attitude system to overcome the influence of time-varying external disturbances. A trajectory tracking control method capable of actively compensating for unknown disturbances in quadrotor UAVs was introduced in [37]. More recently, an active disturbance rejection control (ADRC) technique presented in [38] has attracted significant attention from the control community. Compared to traditional PID control methods, the ADRC technique offers superior performance in disturbance rejection. Moreover, the ADRC method does not require a precise mathematical model of the system, leading to its growing popularity in practical industrial control applications [39], [40], [41], [42]. In recent years, ADRC-based flight control schemes for quadrotor UAVs have been studied. For example, Xu et al. [43] proposed a quadrotor trajectory tracking control scheme that combines backstepping sliding-mode and ADRC techniques together. However, for a swarm of quadrotor UAVs, unknown disturbances impacting each UAV can affect its neighbors via the communication network. Designing a disturbance rejection control scheme for a quadrotor UAV swarm system is inherently more challenging than that for an individual UAV. As far as we know, the disturbance rejection control problem of quadrotor UAV formation has not been well solved, which is the second motivation of this article.

As described in [27], the formation tracking control of a quadrotor UAV swarm system can be decoupled into attitude-loop control and position-loop control. In this article, we design appropriate control schemes for the attitude-loop and position-loop of the quadrotor UAVs. The proposed control scheme has the following characteristics.

1) Different from the time-invariant formation tracking control problem studied in [12], [13], [14], [15], [16], [17], [18], [19], and [20], this article designs a robust formation tracking control scheme for a quadrotor UAV swarm system with time-varying communication topology. In practical applications, the communication topology of the quadrotor UAV swarm may change at any time. Therefore, compared to the results in [12], [13], [14], [15], [16], [17], [18], [19], and [20], the proposed control scheme is more practical.
2) By introducing a distributed adaptive observer to estimate the desired yaw angle, we design a cascade ADRC (CADRC) method for the attitude subsystem. Unlike the results in [17], the quadrotor UAV model considered in this article is not simplified. Therefore, the proposed formation disturbance rejection control method is more valuable in practical applications.
3) The proposed formation control method only requires the position information of each quadrotor UAV. In the existing literature [44], [45], [46], both position and velocity information are required in the design of the formation tracking controller. Compared with these results, the proposed method is also applicable for low-cost quadrotor UAV formations without velocity sensors, and the desired quadrotor formation can be achieved with less interaction information among UAVs.

This article is organized as follows. In Section II, some preliminaries are introduced, and the problem description is presented. In Sections III and IV, a CADRC method and a distributed formation tracking control method are designed for the attitude subsystem and the position subsystem, respectively. The effectiveness of the developed method is verified in Section V. Finally, Section VI concludes this article.

II. Problem Formation

This section presents some basic concepts of graph theory. A mathematical model of quadrotor UAVs is introduced, and the formation tracking control problem of the quadrotor UAV swarm is formulated.

A. Graph Theory

Consider a group of N quadrotor UAVs. If each UAV is regarded as a vertex, the communication topology among UAVs can be described by an undirected graph $\mathcal{G} = (\mathbb{M}, \mathbb{W})$, where $\mathbb{M} \triangleq \{1, 2, \ldots, N\}$ represents the set of vertices, $\mathbb{W} \triangleq \{(i, j) : i \in \mathbb{M}, j \in \mathbb{N}_i\}$ represents the set of edges. For the ith UAV, the neighbor set is defined as $\mathbb{N}_i \triangleq \{j \in \mathbb{M}$: UAV i can receive information from UAV $j\}$. A weight $\mathbf{a}_{ij}$ is assigned to each edge $(i, j) \in \mathbb{W}$, $\mathbf{a}_{ij} = 1$ if $j \in \mathbb{N}_i$ and $\mathbf{a}_{ij} = 0$ otherwise. Then, the Laplacian matrix associated with $\mathcal{G}$ is given as $\mathcal{L} = [\mathbf{w}_{ij}] \in \mathbb{R}^{N\times N}$, where $\mathbf{w}_{ii} = \sum_{j=1, j\neq i}^{N} \mathbf{a}_{ij}$ and $\mathbf{w}_{ij} = -\mathbf{a}_{ij}$ $(i \neq j)$. The undirected graph $\mathcal{G}$ is connected if there is a path between any two vertices. In addition, the leader adjacency matrix is defined as $\mathcal{D} = \mathrm{diag}\{\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_N\}$, where $\mathrm{diag}\{\cdot\}$ represents a diagonal matrix. For the ith UAV, $\mathbf{d}_i > 0$ if it can receive information from the leader and $\mathbf{d}_i = 0$ otherwise.

For the quadrotor UAV swarm system under consideration, the communication topology among UAVs changes over time. Define $\mathbb{G} \triangleq \{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_h\}$ as the set of all possible communication topology graphs, and define $\mathbb{H} \triangleq \{1, 2, \ldots, h\}$ as its index set. Suppose that there is an infinite sequence of time intervals $[t_l, t_{l+1})$ $(l = 1, 2, \ldots)$, where each interval is bounded and nonoverlapping, and $t_0 = 0$. Define a switching signal $\sigma : [0, \infty) \to \mathbb{H}$, which switches at t_l. The matrices $\mathcal{D}_s$ $(s \in \mathbb{H})$ and $\mathcal{L}_s$ associated with the switching communication topology are time varying, but they are time invariant in any time interval $[t_l, t_{l+1})$.

Next, two useful lemmas are introduced.

Lemma 1 [47]: If the undirected graph $\mathcal{G}_s$ $(s \in \mathbb{H})$ is connected, and at least one UAV can receive information from the leader, then the symmetric matrix $\mathcal{H}_s = \mathcal{L}_s + \mathcal{D}_s$ is positive definite.

Lemma 2 [48]: Suppose that a symmetric matrix is partitioned as $\Delta = \begin{bmatrix} \Delta_1 & \Delta_2 \\ \Delta_2^T & \Delta_3 \end{bmatrix}$, the following three conditions are equivalent.

1) $\Delta < 0$.
2) $\Delta_1 < 0$, $\Delta_3 - \Delta_2^T \Delta_1^{-1} \Delta_2 < 0$.
3) $\Delta_3 < 0$, $\Delta_1 - \Delta_2 \Delta_3^{-1} \Delta_2^T < 0$.

B. Modeling of the Quadrotor UAV

As shown in Fig. 1, we establish a space coordinate system to analyze the mathematical model of the quadrotor, where

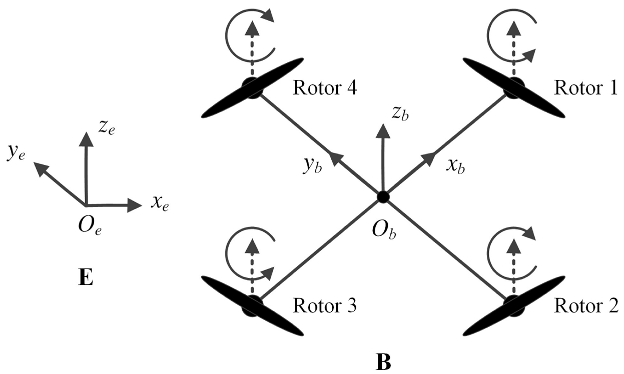

Fig. 1. Configuration of the quadrotor UAV.

E and **B** represent the Earth-fixed frame and the body-fixed frame, respectively. In the body-fixed frame, the quadrotor attitude is defined as $\xi = [\phi, \theta, \varphi]^T \in \mathbb{R}^3$, where ϕ, θ, and φ represent the angles of roll, pitch, and yaw, respectively. In this article, the roll and pitch angles are assumed to vary within the interval $[-\pi/2, \pi/2]$. In addition, the angular velocity with respect to the attitude is defined as $\varpi = [p, q, r]^T \in \mathbb{R}^3$. The rotation matrix between **E** and **B** can be described as

$$\mathcal{R}_t = \begin{bmatrix} C_\theta C_\varphi & S_\theta S_\phi C_\varphi - C_\phi S_\varphi & S_\theta C_\phi C_\varphi + S_\phi S_\varphi \\ C_\theta S_\varphi & S_\theta S_\phi S_\varphi + C_\phi C_\varphi & S_\theta C_\phi S_\varphi - S_\phi C_\varphi \\ -S_\theta & C_\theta S_\phi & C_\theta C_\phi \end{bmatrix} \quad (1)$$

where $S_{(\cdot)}$ and $C_{(\cdot)}$ denote $\sin(\cdot)$ and $\cos(\cdot)$, respectively.

According to the work in [49], the relationship between attitude angle and angular velocity can be expressed as

$$\begin{pmatrix} \dot\phi \\ \dot\theta \\ \dot\varphi \end{pmatrix} = \begin{bmatrix} 1 & T_\theta S_\phi & T_\theta C_\phi \\ 0 & C_\phi & -S_\phi \\ 0 & S_\phi / C_\theta & C_\phi / C_\theta \end{bmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix} \quad (2)$$

where $T_{(\cdot)}$ denotes $\tan(\cdot)$.

By employing the Newton–Euler formulation, the following dynamic equations can be obtained:

$$\mathcal{I}_b \dot\varpi = -\varpi \times \mathcal{I}_b \varpi - \mathcal{T}_g - \mathcal{T}_d + \mathcal{T}_b \quad (3a)$$

$$\mathcal{T}_g = \sum_{i=1}^{4} \mathcal{I}_r (\varpi \times e_3)(-1)^{i+1} \Omega_i \quad (3b)$$

$$\mathcal{T}_d = \operatorname{diag}\{c_\phi, c_\theta, c_\varphi\} \dot\xi \quad (3c)$$

$$\mathcal{T}_b = \begin{bmatrix} lb(\Omega_4^2 - \Omega_2^2) \\ lb(\Omega_3^2 - \Omega_1^2) \\ d(\Omega_1^2 - \Omega_2^2 + \Omega_3^2 - \Omega_4^2) \end{bmatrix} \quad (3d)$$

where $\mathcal{I}_b = \operatorname{diag}\{\mathcal{I}_x, \mathcal{I}_y, \mathcal{I}_z\}$ is a diagonal constant matrix; the notation $\times$ represents cross product; $\mathcal{T}_g$ is the resultant torque induced by the gyroscopic effect; $\mathcal{T}_d$ is the aerodynamic frictions torque; $\mathcal{T}_b$ is the torque provided by the rotors; $e_3 = [0\ 0\ 1]^T \in \mathbb{R}^3$; and the definitions of quadrotor parameters are presented in Table I.

With the aid of (3a)–(3d), the following dynamic equations can be derived:

$$\begin{cases} \dot p = \kappa_1 qr - \kappa_2 \mathcal{M} q - \kappa_3 p + U_p \\ \dot q = \kappa_4 pr + \kappa_5 \mathcal{M} p - \kappa_6 q + U_q \\ \dot r = \kappa_7 pq - \kappa_8 r + U_r \end{cases} \quad (4)$$

TABLE I
QUADROTOR PARAMETERS

Parameter	Description
m	Mass of the quadrotor UAV
g	Acceleration of gravity
l	Distance between center of mass and rotor
b	Lift coefficient
d	Reverse moment coefficient
$\mathcal{I}_r, \mathcal{I}_x, \mathcal{I}_y, \mathcal{I}_z$	Rotary inertia
c_x, c_y, c_z	Drag coefficients
$c_\phi, c_\theta, c_\varphi$	Drag coefficients
$\Omega_1, \Omega_2, \Omega_3, \Omega_4$	Rotary speed of each rotor

where

$$\kappa_1 = \frac{\mathcal{I}_y - \mathcal{I}_z}{\mathcal{I}_x}, \quad \kappa_2 = \frac{\mathcal{I}_r}{\mathcal{I}_x}, \quad \kappa_3 = \frac{c_\phi}{\mathcal{I}_x}, \quad \kappa_4 = \frac{\mathcal{I}_z - \mathcal{I}_x}{\mathcal{I}_y}$$
$$\kappa_5 = \frac{\mathcal{I}_r}{\mathcal{I}_y}, \quad \kappa_6 = \frac{c_\theta}{\mathcal{I}_y}, \quad \kappa_7 = \frac{\mathcal{I}_x - \mathcal{I}_y}{\mathcal{I}_z}, \quad \kappa_8 = \frac{c_\varphi}{\mathcal{I}_z}$$
$$\mathcal{M} = \Omega_1 - \Omega_2 + \Omega_3 - \Omega_4$$
$$U_p = lb(\Omega_4^2 - \Omega_2^2)/\mathcal{I}_x, \quad U_q = lb(\Omega_3^2 - \Omega_1^2)/\mathcal{I}_y$$
$$U_r = d(\Omega_1^2 - \Omega_2^2 + \Omega_3^2 - \Omega_4^2)/\mathcal{I}_z.$$

The position of the quadrotor is defined as $h = [x, y, z]^T \in \mathbb{R}^3$. Then, the translational dynamic equations can be modeled as

$$\begin{bmatrix} \ddot x \\ \ddot y \\ \ddot z \end{bmatrix} = \mathcal{R}_t \begin{bmatrix} 0 \\ 0 \\ U_s \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} - \frac{1}{m} \begin{bmatrix} c_x \dot x \\ c_y \dot y \\ c_z \dot z \end{bmatrix} \quad (5)$$

where $U_s = b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2)/m$.

C. Formation Tracking Control Problem

This article considers the formation tracking control problem of a group of N quadrotor UAVs with switching communication topology $\mathcal{G}_s$ ($s \in \mathbb{H}$). The position dynamic system of each UAV can be extracted as follows:

$$\begin{cases} \dot h_i = v_i \\ \dot v_i = u_i + f_i(v_i), \qquad i = 1, \ldots, N \end{cases} \quad (6)$$

where $h_i = [x_i, y_i, z_i]^T \in \mathbb{R}^3$ and $v_i = [\dot x_i, \dot y_i, \dot z_i]^T \in \mathbb{R}^3$ are the position and velocity, respectively; $u_i = [u_{x_i}, u_{y_i}, u_{z_i}]^T \in \mathbb{R}^3$ is the control input, and

$$u_{x_i} = (C_{\phi_i} S_{\theta_i} C_{\varphi_i} + S_{\phi_i} S_{\varphi_i}) U_{s_i}$$
$$u_{y_i} = (C_{\phi_i} S_{\theta_i} S_{\varphi_i} - S_{\phi_i} C_{\varphi_i}) U_{s_i}$$
$$u_{z_i} = C_{\phi_i} C_{\theta_i} U_{s_i} - g$$
$$f_i(v_i) = \frac{1}{m} [-c_x \dot x_i, -c_y \dot y_i, -c_z \dot z_i]^T.$$

In this article, a virtual leader is introduced to guide the quadrotor UAV swarm to fly along the desired trajectory. The position dynamic system of the virtual leader is given as

$$\dot h_0 = v_0, \ \ \dot v_0 = u_0 \quad (7)$$

where $h_0 = [x_0, y_0, z_0]^T \in \mathbb{R}^3$ and $v_0 = [\dot x_0, \dot y_0, \dot z_0]^T \in \mathbb{R}^3$ are the position and velocity, respectively; and $u_0 = [u_{x_0}, u_{y_0}, u_{z_0}]^T \in \mathbb{R}^3$ is the control input.

In addition, the leader's yaw angle system is described as

$$\dot\varphi_0 = r_0, \quad \dot r_0 = 0 \quad (8)$$

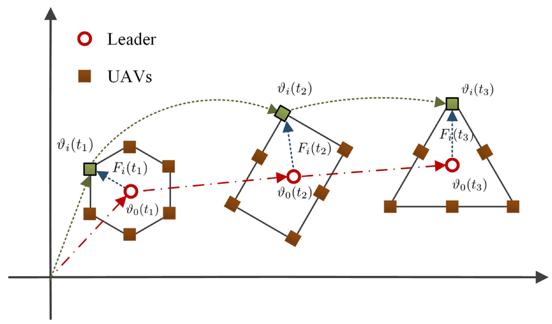

Fig. 2. Time-varying formation of six UAVs.

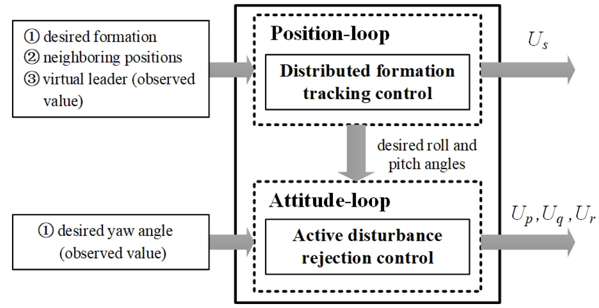

Fig. 3. Control block diagram of the quadrotor UAV.

where φ_0 and r_0 are the yaw angle and yaw angular velocity of the leader, respectively.

Remark 1: In many practical applications, such as plant protection UAVs in agricultural spraying [8], it is usually necessary to set a desired flight trajectory to guide the quadrotor UAV swarm to complete the task. By treating the desired flight trajectory as a virtual leader, the leader-following approach can be used to solve the formation tracking control problem for the quadrotor UAV swarm.

Assumption 1: The communication topology graph $\mathcal{G}_s$ ($s \in \mathbb{H}$) is connected, and at least one UAV can receive information from the virtual leader.

Assumption 2: The control input u_0 satisfies $\|u_0\| \le \bar{u}$, where $\bar{u}$ is an unknown positive constant.

Remark 2: Note that Assumption 1 has been widely used in the existing literature. Assumption 2 must hold in practical applications because all states and control inputs of the quadrotor UAV swarm system are bounded, and the given desired flight trajectory is also bounded.

Definition: A time-varying formation formed by a group of N UAVs is specified by $F = [F_1^T, \ldots, F_N^T]^T \in \mathbb{R}^{6N}$, where $F_i = [F_{h_i}^T, F_{v_i}^T]^T \in \mathbb{R}^6$ is the formation vector of UAV i, and it is piecewise continuously differentiable. Then, the quadrotor UAV swarm system achieves time-varying formation tracking flight if the following condition is satisfied:

$$\lim_{t\to+\infty} (\vartheta_i(t) - F_i(t) - \vartheta_0(t)) = 0, \quad i = 1, \ldots, N \tag{9}$$

where $\vartheta_i = [h_i^T, v_i^T]^T$ and $\vartheta_0 = [h_0^T, v_0^T]^T$.

To illustrate the relationship between ϑ_i, F_i, and ϑ_0, a formation of six UAVs on a 2-D plane is shown in Fig. 2.

Control Objectives: The control for each quadrotor UAV is divided into two steps: 1) designing a CADRC method for the attitude subsystem and 2) designing a distributed formation tracking control method for the position subsystem. The control block diagram of the quadrotor UAV is shown in Fig. 3. This article achieves the following control objectives for the quadrotor UAV swarm: 1) forming a desired formation; 2) tracking the desired flight trajectory; and 3) reducing the influence of disturbances.

III. Active Disturbance Rejection Control of the Attitude Subsystem

In this section, we will design a CADRC method for the attitude subsystem. This method aims to ensure the stability of the attitude subsystem even in the presence of unknown disturbances.

A. Distributed Observer Design

Note that the desired yaw angle φ_0 cannot be measured unless there is a communication link between the UAV and the virtual leader. Therefore, φ_0 cannot be used directly in the design of the attitude controller. In this article, a distributed observer is designed for each UAV to estimate φ_0. This design process includes the following two steps.

Step 1: A neighbor-based system to estimate r_0

$$\dot{\hat{r}}_i = -\gamma \ell \Phi_i \tag{10}$$

where $\hat{r}_i$ is the estimation of r_0; ℓ and γ are positive constants with $\gamma < 1$; and $\Phi_i = \sum_{j\in\mathbb{N}_i(t)} \mathbf{a}_{ij}(\hat{\varphi}_i - \hat{\varphi}_j) + \mathbf{d}_i(\hat{\varphi}_i - \varphi_0)$.

Step 2: A neighbor-based observer to estimate φ_0

$$\dot{\hat{\varphi}}_i = -\ell \Phi_i + \hat{r}_i \tag{11}$$

where $\hat{\varphi}_i$ is the estimation of φ_0.

Define $\hat{\varphi} = [\hat{\varphi}_1, \ldots, \hat{\varphi}_N]^T$ and $\hat{r} = [\hat{r}_1, \ldots, \hat{r}_N]^T$, the following compact form can be obtained:

$$\begin{cases} \dot{\hat{\varphi}} = -\ell \mathcal{H}_\sigma \otimes \hat{\varphi} + \ell \mathcal{D}_\sigma \mathbf{1}_N \otimes \varphi_0 + \hat{r} \\ \dot{\hat{r}} = -\gamma \ell \mathcal{H}_\sigma \otimes \hat{\varphi} + \gamma \ell \mathcal{D}_\sigma \mathbf{1}_N \otimes \varphi_0 \end{cases} \tag{12}$$

where $\mathcal{H}_\sigma = \mathcal{L}_\sigma + \mathcal{D}_\sigma$ is a positive-definite symmetric matrix; $\sigma : [0, \infty) \to \mathbb{H}$ is a switching signal; $\otimes$ denotes the Kronecker product; and $\mathbf{1}_N$ represents a vector with all entries equal to 1.

Next, define $\bar{\varphi} = \hat{\varphi} - \mathbf{1}_N \otimes \varphi_0$ and $\bar{r} = \hat{r} - \mathbf{1}_N \otimes r_0$. Considering that $-\ell \mathcal{H}_\sigma \otimes \hat{\varphi} + \ell \mathcal{D}_\sigma \mathbf{1}_N \otimes \varphi_0 = -\ell \mathcal{H}_\sigma \otimes \bar{\varphi}$. Then, one can obtain a dynamic error system as follows:

$$\dot{e} = J_\sigma e \tag{13}$$

where $e = \begin{bmatrix} \bar{\varphi} \\ \bar{r} \end{bmatrix}$ and $J_\sigma = \begin{bmatrix} -\ell \mathcal{H}_\sigma & I_N \\ -\gamma \ell \mathcal{H}_\sigma & 0 \end{bmatrix} \otimes I_2$.

Now, we present the following theorem.

Theorem 1: For any given positive constants $\gamma < 1$ and $\ell > (1/[4\gamma(1-\gamma^2)\lambda_{\min}])$, where $\lambda_{\min}$ represents the minimum eigenvalue of the matrix $\mathcal{H}_s$ ($s \in \mathbb{H}$). If the switching communication topology keeps connected, then

$$\lim_{t\to+\infty} \|e(t)\| = 0. \tag{14}$$

Proof: We consider a Lyapunov function $V(e(t)) = e^T(t) X e(t)$, where $X = \begin{bmatrix} I_N & -\gamma I_N \\ -\gamma I_N & I_N \end{bmatrix} \otimes I_2$ is a symmetric positive-definite matrix. In the following, the time variable t is omitted for convenience of description.

The communication topology is time varying, but the matrix J_s ($s \in \mathbb{H}$) associated with the communication topology is time invariant in each time interval $[t_l, t_{l+1})$ ($l = 1, 2, \ldots$). From (13), the derivative of $V(e)$ satisfies

$$\dot{V}(e) = e^T\big(J_s^T X + XJ_s\big)e = e^T Ye \tag{15}$$

where $Y = \begin{bmatrix} -2\ell(1-\gamma^2)\mathcal{H}_s & I_N \\ I_N & -2\gamma I_N \end{bmatrix} \otimes I_2$.

Note that $-2\ell(1-\gamma^2)\mathcal{H}_s < 0$ and $-2\gamma I_N + 1/(2\ell(1-\gamma^2))\mathcal{H}_s^{-1} < 0$, from Lemma 2, it follows that $Y < 0$. Then, Theorem 1 can be obtained. This completes the proof. ■

Remark 3: The proposed distributed observer can be used to estimate φ_0, and the estimation error eventually converges to zero. Since φ_0 is bounded, then $\hat{\varphi}_i$ is also bounded.

B. Attitude Controller Design

For convenience of analysis, the three attitude angles of each UAV are described by a unified attitude angle system. For the ith UAV, define $m_{i,1} = \phi_i$, $m_{i,2} = \theta_i$, $m_{i,3} = \varphi_i$, $n_{i,1} = p_i$, $n_{i,2} = q_i$, and $n_{i,3} = r_i$, then the following unified cascade attitude angle system can be obtained:

$$\begin{cases} \mathcal{S}_{m_{i,j}} : \dot{m}_{i,j} = n_{i,j} + f_{m_{i,j}} + w_{m_{i,j}} \\ \mathcal{S}_{n_{i,j}} : \dot{n}_{i,j} = U_{n_{i,j}} + f_{n_{i,j}} + w_{n_{i,j}}, \quad j = 1, 2, 3 \end{cases} \tag{16}$$

where $w_{m_{i,j}}$ and $w_{n_{i,j}}$ represent some unknown time-varying disturbances, and

$$\begin{aligned} f_{m_{i,1}} &= T_{m_{i,2}} S_{m_{i,1}} n_{i,2} + T_{m_{i,2}} C_{m_{i,1}} n_{i,3} \\ f_{m_{i,2}} &= C_{m_{i,1}} n_{i,2} - S_{m_{i,1}} n_{i,3} - n_{i,2} \\ f_{m_{i,3}} &= S_{m_{i,1}}/C_{m_{i,2}} n_{i,2} + C_{m_{i,1}}/C_{m_{i,2}} n_{i,3} - n_{i,3} \\ f_{n_{i,1}} &= \kappa_1 n_{i,2} n_{i,3} - \kappa_2 \mathcal{M}_i n_{i,2} - \kappa_3 n_{i,1} \\ f_{n_{i,2}} &= \kappa_4 n_{i,1} n_{i,3} + \kappa_5 \mathcal{M}_i n_{i,1} - \kappa_6 n_{i,2} \\ f_{n_{i,3}} &= \kappa_7 n_{i,1} n_{i,2} - \kappa_8 n_{i,3}. \end{aligned}$$

Assumption 3: The unknown disturbances $w_{m_{i,j}}$ and $w_{n_{i,j}}$ ($i = 1, \ldots, N; j = 1, 2, 3$) satisfy

$$|\dot{w}_{m_{i,j}}| \le \bar{w}_m, \quad |\dot{w}_{n_{i,j}}| \le \bar{w}_n \tag{17}$$

where $\bar{w}_m$ and $\bar{w}_n$ are unknown positive constants.

Remark 4: Assumption 3 is reasonable since the time-varying disturbances in actual systems are usually bounded.

In what follows, active disturbance rejection controllers are designed for subsystems $\mathcal{S}_{m_{i,j}}$ and $\mathcal{S}_{n_{i,j}}$. Define $\phi_{i,0}$ as the desired roll angle and $\theta_{i,0}$ as the desired pitch angle. Then, similar to the backstepping technique, define $u_{m_{i,j}} = n_{i,j}$ as pseudo-control variables. The controller design process includes the following two steps.

Step 1: For subsystem $\mathcal{S}_{m_{i,j}}$, the tracking error is defined as $\alpha_{i,j,1} = m_{i,j} - \hbar_{i,j}$, where $\hbar_{i,1} = \phi_{i,0}$, $\hbar_{i,2} = \theta_{i,0}$, and $\hbar_{i,3} = \hat{\varphi}_i$. It can be obtained from (16) that

$$\dot{\alpha}_{i,j,1} = f_{m_{i,j}} + u_{m_{i,j}} + w_{m_{i,j}} - \dot{\hbar}_{i,j}. \tag{18}$$

Define $w_{m_{i,j}}$ as an extended state variable $\alpha_{i,j,2}$. Then, one can obtain that

$$\begin{cases} \dot{\alpha}_{i,j,1} = f_{m_{i,j}} + u_{m_{i,j}} + \alpha_{i,j,2} - \dot{\hbar}_{i,j} \\ \dot{\alpha}_{i,j,2} = \dot{w}_{m_{i,j}}. \end{cases} \tag{19}$$

Next, an extended state observer (ESO) is designed to estimate the tracking error

$$\begin{cases} \dot{\hat{\alpha}}_{i,j,1} = \hat{\alpha}_{i,j,2} + \mathcal{Q}_{i,1}\big(\tilde{\alpha}_{i,j,1}\big) + f_{m_{i,j}} + u_{m_{i,j}} - \dot{\hbar}_{i,j} \\ \dot{\hat{\alpha}}_{i,j,2} = \lambda_{i,j}^{-1} \mathcal{Q}_{i,2}\big(\tilde{\alpha}_{i,j,1}\big) \end{cases} \tag{20}$$

where $\hat{\alpha}_{i,j} = [\hat{\alpha}_{i,j,1}, \hat{\alpha}_{i,j,2}]^T \in \mathbb{R}^2$ is the observer state; $\tilde{\alpha}_{i,j,1} = ([\alpha_{i,j,1} - \hat{\alpha}_{i,j,1}]/\lambda_{i,j})$, and $\lambda_{i,j}$ ($0 < \lambda_{i,j} < 1$) is a design parameter. In addition, the functions $\mathcal{Q}_{i,1}(\cdot)$ and $\mathcal{Q}_{i,2}(\cdot)$ are chosen as

$$\mathcal{Q}_{i,1}(\nu_i) = b_{i,1}\nu_i, \quad \mathcal{Q}_{i,2}(\nu_i) = b_{i,2}\nu_i \tag{21}$$

where $b_{i,1}$ and $b_{i,2}$ are positive constants such that the matrix $B_i = \begin{bmatrix} -b_{i,1} & 1 \\ -b_{i,2} & 0 \end{bmatrix}$ is a Hurwitz matrix.

Define $\tilde{\alpha}_{i,j,2} = \alpha_{i,j,2} - \hat{\alpha}_{i,j,2}$, it follows from (19) and (20) that:

$$\begin{cases} \dot{\tilde{\alpha}}_{i,j,1} = \lambda_{i,j}^{-1}\big[\tilde{\alpha}_{i,j,2} - \mathcal{Q}_{i,1}\big(\tilde{\alpha}_{i,j,1}\big)\big] \\ \dot{\tilde{\alpha}}_{i,j,2} = \dot{\alpha}_{i,j,2} - \lambda_{i,j}^{-1} \mathcal{Q}_{i,2}\big(\tilde{\alpha}_{i,j,1}\big) \end{cases} \tag{22}$$

where $\tilde{\alpha}_{i,j} = [\tilde{\alpha}_{i,j,1}, \tilde{\alpha}_{i,j,2}]^T$ is the estimation error.

The pseudo-control signal $u_{m_{i,j}}$ is designed as

$$u_{m_{i,j}} = -k_{m_{i,j}} \lambda_{i,j}^{-1} \hat{\alpha}_{i,j,1} - \hat{\alpha}_{i,j,2} - f_{m_{i,j}} + \dot{\hbar}_{i,j} \tag{23}$$

where $k_{m_{i,j}}$ is a positive design parameter.

Step 2: For subsystem $\mathcal{S}_{n_{i,j}}$, the tracking error is defined as $\beta_{i,j,1} = n_{i,j} - u_{m_{i,j}}$. It can be obtained from (16) that

$$\dot{\beta}_{i,j,1} = f_{n_{i,j}} + U_{n_{i,j}} + w_{n_{i,j}} - \dot{u}_{m_{i,j}}. \tag{24}$$

Define $w_{n_{i,j}}$ as an extended state variable $\beta_{i,j,2}$. Then, one can obtain that

$$\begin{cases} \dot{\beta}_{i,j,1} = f_{n_{i,j}} + U_{n_{i,j}} + \beta_{i,j,2} - \dot{u}_{m_{i,j}} \\ \dot{\beta}_{i,j,2} = \dot{w}_{n_{i,j}}. \end{cases} \tag{25}$$

Next, an ESO is introduced to estimate the tracking error

$$\begin{cases} \dot{\hat{\beta}}_{i,j,1} = \hat{\beta}_{i,j,2} + \mathcal{Q}_{i,1}\Big(\tilde{\beta}_{i,j,1}\Big) + f_{n_{i,j}} + U_{n_{i,j}} - \dot{u}_{m_{i,j}} \\ \dot{\hat{\beta}}_{i,j,2} = \lambda_{i,j}^{-1} \mathcal{Q}_{i,2}\Big(\tilde{\beta}_{i,j,1}\Big) \end{cases} \tag{26}$$

where $\hat{\beta}_{i,j} = [\hat{\beta}_{i,j,1}, \hat{\beta}_{i,j,2}]^T \in \mathbb{R}^2$ is the observer state; and $\tilde{\beta}_{i,j,1} = ([\beta_{i,j,1} - \hat{\beta}_{i,j,1}]/[\lambda_{i,j}])$.

Define $\tilde{\beta}_{i,j,2} = \beta_{i,j,2} - \hat{\beta}_{i,j,2}$, it follows from (25) and (26) that:

$$\begin{cases} \dot{\tilde{\beta}}_{i,j,1} = \lambda_{i,j}^{-1}\Big[\tilde{\beta}_{i,j,2} - \mathcal{Q}_{i,1}\Big(\tilde{\beta}_{i,j,1}\Big)\Big] \\ \dot{\tilde{\beta}}_{i,j,2} = \dot{\beta}_{i,j,2} - \lambda_{i,j}^{-1} \mathcal{Q}_{i,2}\Big(\tilde{\beta}_{i,j,1}\Big) \end{cases} \tag{27}$$

where $\tilde{\beta}_{i,j} = [\tilde{\beta}_{i,j,1}, \tilde{\beta}_{i,j,2}]^T \in \mathbb{R}^2$ is the estimation error.

The controller $U_{n_{i,j}}$ is designed as

$$U_{n_{i,j}} = -k_{n_{i,j}} \lambda_{i,j}^{-1} \hat{\beta}_{i,j,1} - \hat{\beta}_{i,j,2} - f_{n_{i,j}} + \dot{u}_{m_{i,j}} \tag{28}$$

where $k_{n_{i,j}}$ is a positive design parameter.

Remark 5: Note that the desired roll angle $\phi_{i,0}$ and the desired pitch angle $\theta_{i,0}$ are generated by the position loop of each quadrotor UAV. Therefore, there is no need to design additional distributed observers for each quadrotor UAV to estimate $\phi_{i,0}$ and $\theta_{i,0}$.

C. Stability Analysis

Now, we present the analysis results.

Theorem 2: Consider the attitude angle system in (16), the observers in (20) and (26), and the controller in (28). There exists a constant $\lambda_{i,j}^*$ $(0 < \lambda_{i,j}^* < 1)$ such that for any $\lambda_{i,j} \in (0, \lambda_{i,j}^*)$, all states of the closed-loop system (16) are globally bounded. By adjusting the design parameter $\lambda_{i,j}$, the tracking error of the attitude angle system can converge to an arbitrary small set.

Proof: Define a function $\mathcal{P}_i(\nu_i) : \mathbb{R}^2 \to \mathbb{R}$ as

$$\mathcal{P}_i(\nu_i) = \nu_i^T P_i \nu_i \tag{29}$$

where the positive-definite matrix P_i satisfies $P_i B_i + B_i^T P_i = -I_2$. Inspired by [50], by choosing functions $\mathcal{Q}_{i,1}(\cdot)$, $\mathcal{Q}_{i,2}(\cdot)$, and $\mathcal{P}_i(\cdot)$ in the form of (21) and (29), one can find a function $\Upsilon_i(\cdot) : \mathbb{R}^2 \to \mathbb{R}$ such that

$$\varepsilon_1 \|\nu_i\|^2 \le \mathcal{P}_i(\nu_i) \le \varepsilon_2 \|\nu_i\|^2 \tag{30}$$

$$\varepsilon_3 \|\nu_i\|^2 \le \Upsilon_i(\nu_i) \le \varepsilon_4 \|\nu_i\|^2 \tag{31}$$

$$\begin{aligned} &\frac{\partial \mathcal{P}_i(\nu_i)}{\partial \nu_{i,1}}\left[\nu_{i,2} - \mathcal{Q}_{i,1}(\nu_{i,1})\right] - \frac{\partial \mathcal{P}_i(\nu_i)}{\partial \nu_{i,2}} \mathcal{Q}_{i,2}(\nu_{i,1}) \\ &\quad \le -\Upsilon_i(\nu_i) \end{aligned} \tag{32}$$

$$\left\| \frac{\partial \mathcal{P}_i(\nu_i)}{\partial \nu_{i,l}} \right\| \le \varepsilon_5 \|\nu_i\|, \quad l = 1, 2 \tag{33}$$

where $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4$, and ε_5 are some positive constants.

We consider a Lyapunov function

$$V_{i,j} = \alpha_{i,j,1}^2 + \beta_{i,j,1}^2 + \mathcal{P}_i(\tilde{\alpha}_{i,j}) + \mathcal{P}_i(\tilde{\beta}_{i,j}). \tag{34}$$

The derivative of (34) satisfies the following inequality:

$$\begin{aligned} \dot{V}_{i,j} \le\; & -2k_{m_{i,j}}\lambda_{i,j}^{-1}\alpha_{i,j,1}^2 + 2\alpha_{i,j,1}\left(\tilde{\alpha}_{i,j,2} + k_{m_{i,j}}\tilde{\alpha}_{i,j,1}\right) \\ & - 2k_{n_{i,j}}\lambda_{i,j}^{-1}\beta_{i,j,1}^2 + 2\beta_{i,j,1}\left(\tilde{\beta}_{i,j,2} + k_{n_{i,j}}\tilde{\beta}_{i,j,1}\right) \\ & + \lambda_{i,j}^{-1}\left\{ \frac{\partial \mathcal{P}_i(\tilde{\alpha}_{i,j})}{\partial \tilde{\alpha}_{i,j,1}}\left[\tilde{\alpha}_{i,j,2} - \mathcal{Q}_{i,1}(\tilde{\alpha}_{i,j,1})\right] \right. \\ & \left. - \frac{\partial \mathcal{P}_i(\tilde{\alpha}_{i,j})}{\partial \tilde{\alpha}_{i,j,2}} \mathcal{Q}_{i,2}(\tilde{\alpha}_{i,j,1}) \right\} + \left| \frac{\partial \mathcal{P}_i(\tilde{\alpha}_{i,j})}{\partial \tilde{\alpha}_{i,j,2}} \right| |\dot{\alpha}_{i,j,2}| \\ & + \lambda_{i,j}^{-1}\left\{ \frac{\partial \mathcal{P}_i(\tilde{\beta}_{i,j})}{\partial \tilde{\beta}_{i,j,1}}\left[\tilde{\beta}_{i,j,2} - \mathcal{Q}_{i,1}(\tilde{\beta}_{i,j,1})\right] \right. \\ & \left. - \frac{\partial \mathcal{P}_i(\tilde{\beta}_{i,j})}{\partial \tilde{\beta}_{i,j,2}} \mathcal{Q}_{i,2}(\tilde{\beta}_{i,j,1}) \right\} + \left| \frac{\partial \mathcal{P}_i(\tilde{\beta}_{i,j})}{\partial \tilde{\beta}_{i,j,2}} \right| |\dot{\beta}_{i,j,2}|. \end{aligned} \tag{35}$$

Applying Young's inequality, the following inequalities can be obtained:

$$\begin{aligned} &2\alpha_{i,j,1}\left(\tilde{\alpha}_{i,j,2} + k_{m_{i,j}}\tilde{\alpha}_{i,j,1}\right) \\ &\quad \le \left(1 + k_{m_{i,j}}\right)\left(\alpha_{i,j,1}^2 + \|\tilde{\alpha}_{i,j}\|^2\right) \end{aligned} \tag{36}$$

$$\begin{aligned} &2\beta_{i,j,1}\left(\tilde{\beta}_{i,j,2} + k_{n_{i,j}}\tilde{\beta}_{i,j,1}\right) \\ &\quad \le \left(1 + k_{n_{i,j}}\right)\left(\beta_{i,j,1}^2 + \|\tilde{\beta}_{i,j}\|^2\right). \end{aligned} \tag{37}$$

Considering (31) and (32), it can be proved that

$$\begin{aligned} &\frac{\partial \mathcal{P}_i(\tilde{\alpha}_{i,j})}{\partial \tilde{\alpha}_{i,j,1}}\left[\tilde{\alpha}_{i,j,2} - \mathcal{Q}_{i,1}(\tilde{\alpha}_{i,j,1})\right] - \frac{\partial \mathcal{P}_i(\tilde{\alpha}_{i,j})}{\partial \tilde{\alpha}_{i,j,2}} \mathcal{Q}_{i,2}(\tilde{\alpha}_{i,j,1}) \\ &\quad \le -\Upsilon_i(\tilde{\alpha}_{i,j}) \le -\varepsilon_3 \|\tilde{\alpha}_{i,j}\|^2 \end{aligned} \tag{38}$$

$$\begin{aligned} &\frac{\partial \mathcal{P}_i(\tilde{\beta}_{i,j})}{\partial \tilde{\beta}_{i,j,1}}\left[\tilde{\beta}_{i,j,2} - \mathcal{Q}_{i,1}(\tilde{\beta}_{i,j,1})\right] - \frac{\partial \mathcal{P}_i(\tilde{\beta}_{i,j})}{\partial \tilde{\beta}_{i,j,2}} \mathcal{Q}_{i,2}(\tilde{\beta}_{i,j,1}) \\ &\quad \le -\Upsilon_i(\tilde{\beta}_{i,j}) \le -\varepsilon_3 \|\tilde{\beta}_{i,j}\|^2. \end{aligned} \tag{39}$$

Moreover, from Assumption 3 and (33), one has

$$\begin{aligned} \left| \frac{\partial \mathcal{P}_i(\tilde{\alpha}_{i,j})}{\partial \tilde{\alpha}_{i,j,2}} \right| |\dot{\alpha}_{i,j,2}| &= \left| \frac{\partial \mathcal{P}_i(\tilde{\alpha}_{i,j})}{\partial \tilde{\alpha}_{i,j,2}} \right| |\dot{w}_{m_{i,j}}| \\ &\le \|\tilde{\alpha}_{i,j}\|^2 + \frac{1}{4}\varepsilon_5^2 \bar{w}_m^2 \end{aligned} \tag{40}$$

$$\begin{aligned} \left| \frac{\partial \mathcal{P}_i(\tilde{\beta}_{i,j})}{\partial \tilde{\beta}_{i,j,2}} \right| |\dot{\beta}_{i,j,2}| &= \left| \frac{\partial \mathcal{P}_i(\tilde{\beta}_{i,j})}{\partial \tilde{\beta}_{i,j,2}} \right| |\dot{w}_{n_{i,j}}| \\ &\le \|\tilde{\beta}_{i,j}\|^2 + \frac{1}{4}\varepsilon_5^2 \bar{w}_n^2. \end{aligned} \tag{41}$$

According to (35)–(41), it follows that:

$$\begin{aligned} \dot{V}_{i,j} \le\; & -\Xi_{i,j,1}\alpha_{i,j,1}^2 - \Xi_{i,j,2}\beta_{i,j,1}^2 - \Xi_{i,j,3}\|\tilde{\alpha}_{i,j}\|^2 \\ & - \Xi_{i,j,4}\|\tilde{\beta}_{i,j}\|^2 + \Pi_{i,j} \end{aligned} \tag{42}$$

where

$$\begin{aligned} \Xi_{i,j,1} &= 2k_{m_{i,j}}\lambda_{i,j}^{-1} - k_{m_{i,j}} - 1 \\ \Xi_{i,j,2} &= 2k_{n_{i,j}}\lambda_{i,j}^{-1} - k_{n_{i,j}} - 1 \\ \Xi_{i,j,3} &= \varepsilon_3\lambda_{i,j}^{-1} - k_{m_{i,j}} - 2 \\ \Xi_{i,j,4} &= \varepsilon_3\lambda_{i,j}^{-1} - k_{n_{i,j}} - 2 \\ \Pi_{i,j} &= \frac{1}{4}\varepsilon_5^2 \bar{w}_m^2 + \frac{1}{4}\varepsilon_5^2 \bar{w}_n^2. \end{aligned}$$

Now, choose $\lambda_{i,j}$ as follows:

$$0 < \lambda_{i,j} < \lambda_{i,j}^* \tag{43}$$

where $\lambda_{i,j}^* = \min\{(2k_{m_{i,j}}/[k_{m_{i,j}} + 1]), (2k_{n_{i,j}}/[k_{n_{i,j}} + 1]), (\varepsilon_3/[k_{m_{i,j}} + 2]), (\varepsilon_3/[k_{n_{i,j}} + 2]), 1\}$.

Then, the following inequality can be achieved:

$$\dot{V}_{i,j} \le -\delta_{i,j}V_{i,j} + \Pi_{i,j} \tag{44}$$

where $\delta_{i,j} = \min\{\Xi_{i,j,1}, \Xi_{i,j,2}, (\Xi_{i,j,3}/\varepsilon_2), (\Xi_{i,j,4}/\varepsilon_2)\}$.

It follows from (44) that:

$$0 \le V_{i,j}(t) \le \frac{\Pi_{i,j}}{\delta_{i,j}} + \left[V_{i,j}(0) - \frac{\Pi_{i,j}}{\delta_{i,j}}\right] e^{-\delta_{i,j}t}. \tag{45}$$

Thus

$$\lim_{t\to+\infty} |\alpha_{i,j,1}| \le \lim_{t\to+\infty} \sqrt{V_{i,j}(t)} \le \sqrt{\frac{\Pi_{i,j}}{\delta_{i,j}}}. \tag{46}$$

Since $\delta_{i,j}$ increases monotonically with the decrease of $\lambda_{i,j}$. From (46), one can obtain that the tracking error $\alpha_{i,j,1}$ can converge to be arbitrarily small by decreasing $\lambda_{i,j}$. In addition, from (45), it can be proved that all states are globally bounded. This completes the proof. ∎

Remark 6: Different from the ADRC method in [43], this article develops a distributed ADRC method for the quadrotor UAV swarm system by introducing a distributed observer (13) to estimate the desired yaw angle. In addition, if $w_{m_{i,j}} =$

$w_{n_{i,j}} = 0$, one can get that $\Pi_{i,j} = 0$. Then, according to (44), the tracking error of the attitude angle system can converge to zero.

IV. Distributed Formation Tracking Control of the Position Subsystem

In this section, a new distributed formation tracking control method is developed for the position subsystem. This method aims to enable the quadrotor UAV swarm to achieve time-varying formation flying and track a virtual leader.

We first design an observer for each quadrotor UAV to estimate the velocity v_i, the dynamics of the observer are given as follows:

$$\begin{cases} \dot{\hat{v}}_i = S_i + \mu\big(h_i - F_{h_i}\big) \\ \dot{S}_i = u_i + f_i\big(\hat{v}_i\big) - \mu\big(\hat{v}_i - F_{v_i}\big) \end{cases} \tag{47}$$

where $\hat{v}_i \in \mathbb{R}^3$ is the estimation of v_i; and μ is a positive design parameter.

Remark 7: Note that the formation vector $F_i = [F_{h_i}^T, F_{v_i}^T]^T$ is used in the observer (47). The proposed observer is beneficial for the quadrotor UAV swarm to achieve formation flying.

Then, we propose the following formation control protocol.

1) *Distributed Formation Tracking Controller:*

$$u_i = -c\psi_i - \epsilon_i\chi_i - f_i\big(\hat{v}_i\big) + \dot{F}_{v_i} \tag{48}$$

$$\psi_i = \sum_{j\in\mathbb{N}_i(t)} \mathbf{a}_{ij}\big(G_i - G_j\big) + \mathbf{d}_i G_i \tag{49}$$

$$G_i = \frac{\chi_i\hat{\eta}_i^2}{\|\chi_i\|\hat{\eta}_i + \delta_i}. \tag{50}$$

2) *Adaptive Laws:*

$$\dot{\epsilon}_i = \varrho_i\chi_i^T\chi_i \tag{51}$$

$$\dot{\hat{\eta}}_i = \tau_i\|\chi_i\| - \tau_i\hat{\eta}_i\delta_i \tag{52}$$

where c, ϱ_i, and τ_i are positive constants; $\|\cdot\|$ represents the Euclidean norm of a vector; and δ_i is a positive continuous function which satisfies $\lim_{t\to\infty}\int_{t_0}^{t}\delta_i(s)ds \le \bar{\delta}_i < +\infty$, where $\bar{\delta}_i$ is a positive constant. In addition, $\chi_i = \chi_{h_i} + \chi_{v_i}$, and

$$\chi_{h_i} = \sum_{j\in\mathbb{N}_i(t)} \mathbf{a}_{ij}(h_i - F_{h_i} - h_j + F_{h_j}) + \mathbf{d}_i(h_i - F_{h_i} - h_0)$$

$$\chi_{v_i} = \sum_{j\in\mathbb{N}_i(t)} \mathbf{a}_{ij}(\hat{v}_i - F_{v_i} - \hat{v}_j + F_{v_j}) + \mathbf{d}_i(\hat{v}_i - F_{v_i} - v_0).$$

Remark 8: In the distributed formation tracking controller (48), ψ_i is designed to overcome the influence of the bounded input signal u_0, and ϵ_i is an adaptive coupling weight designed to avoid using the global information of the communication topology graph in the formation controller (48).

Next, the formation tracking control performance of the quadrotor UAV swarm is analyzed. Define $\xi_{h_i} = h_i - F_{h_i} - h_0$, $\xi_{v_i} = \hat{v}_i - F_{v_i} - v_0$, and $\zeta_i = v_i - \hat{v}_i$. The following dynamic error system can be obtained:

$$\begin{cases} \dot{\xi}_{h_i} = \xi_{v_i} + \zeta_i \\ \dot{\xi}_{v_i} = \mu\zeta_i - c\psi_i - \epsilon_i\chi_i - u_0 \\ \dot{\zeta}_i = -\mu\zeta_i + f_i(v_i) - f_i\big(\hat{v}_i\big). \end{cases} \tag{53}$$

We define $\xi_h = [\xi_{h_1}, \ldots, \xi_{h_N}]^T$, $\xi_v = [\xi_{v_1}, \ldots, \xi_{v_N}]^T$, $\zeta = [\zeta_1, \ldots, \zeta_N]^T$, $\mathbf{G} = [G_1, \ldots, G_N]^T$, $\chi = [\chi_1, \ldots, \chi_N]^T$, and $\Gamma = [\xi_h, \xi_v, \zeta]^T$. Then, the dynamic error system (53) can be rewritten in a compact form

$$\begin{cases} \dot{\Gamma} = (W_\sigma \otimes I_3)\Gamma + \mathbf{U} + \Theta \\ \Psi = (\mathcal{H}_\sigma \otimes I_3)\mathbf{G} \\ \chi = (\mathcal{H}_\sigma \otimes I_3)(\xi_h + \xi_v) \end{cases} \tag{54}$$

where $\mathcal{H}_\sigma = \mathcal{L}_\sigma + \mathcal{D}_\sigma$ is a positive-definite symmetric matrix, $\sigma : [0, \infty) \to \mathbb{H}$ is a switching signal, and

$$W_\sigma = \begin{bmatrix} 0 & I_N & I_N \\ -\epsilon\mathcal{H}_\sigma & -\epsilon\mathcal{H}_\sigma & \mu I_N \\ 0 & 0 & -\mu I_N \end{bmatrix}$$

$$\mathbf{U} = [0, -c\Psi^T - (\mathbf{1}_N \otimes u_0)^T, 0]^T$$

$$\epsilon = [\epsilon_1, \ldots, \epsilon_N]^T, \quad \Psi = [\psi_1, \ldots, \psi_N]^T$$

$$\Theta = [0, 0, (f(v) - f(\hat{v}))^T]^T$$

$$f(v) = [f_1(v_1), \ldots, f_N(v_N)]^T$$

$$f(\hat{v}) = [f_1(\hat{v}_1), \ldots, f_N(\hat{v}_N)]^T.$$

Note that if Γ asymptotically converges to zero, distributed time-varying formation tracking control of the quadrotor UAV swarm can be achieved.

Remark 9: In this article, only the position information is used in the design of the distributed formation controller. If the velocity information can be obtained directly, formation tracking control of the quadrotor UAV swarm can also be achieved by replacing $\hat{v}_i$ with v_i in the distributed formation controller (48) and introducing some manipulations. Compared with the formation control schemes in [44], [45], and [46], the proposed method is still applicable even if the quadrotor UAV swarm is not equipped with velocity sensors.

Now, we present the analysis results.

Theorem 3: Consider the overall closed-loop system consisting of the quadrotor UAV swarm system (6), the observer (47), the formation tracking controller (48), and the adaptive laws (51) and (52). All states of the closed-loop system are globally bounded, and the quadrotor UAV swarm can achieve time-varying formation flying and track a virtual leader, that is

$$\lim_{t\to+\infty}(\vartheta_i(t) - F_i(t) - \vartheta_0(t)) = 0, \quad i = 1, \ldots, N. \tag{55}$$

Proof: Construct the following Lyapunov function:

$$V_h = \Gamma^T(M \otimes I_3)\Gamma + \frac{1}{2}\sum_{i=1}^{N}\varrho_i^{-1}(\epsilon_i - \epsilon_0)^2 + \rho c\sum_{i=1}^{N}\tau_i^{-1}\tilde{\eta}_i^2 \tag{56}$$

where $M = \begin{bmatrix} 2\rho I_N & \rho I_N & 0 \\ \rho I_N & \rho I_N & 0 \\ 0 & 0 & \bar{\rho} I_N \end{bmatrix}$ is a positive-definite matrix; $\rho = \check{\lambda}/\bar{\lambda}$ with $\bar{\lambda} = \lambda_{\min}(\mathcal{H}_\sigma^T + \mathcal{H}_\sigma)$ and $\check{\lambda} = \lambda_{\max}(\mathcal{H}_\sigma^T\mathcal{H}_\sigma)$; $\tilde{\eta}_i = \hat{\eta}_i - \eta_i$, and η_i is a positive constant to be determined later; and ϵ_0 and $\bar{\rho}$ are positive constants.

The derivative of V_h is derived as

$$\dot{V}_h = 2\Gamma^T(M \otimes I_3)[(W_\sigma \otimes I_3)\Gamma + \mathbf{U} + \Theta]$$

$$+\sum_{i=1}^{N}\varrho_i^{-1}(\epsilon_i-\epsilon_0)\dot{\epsilon}_i+2\rho c\sum_{i=1}^{N}\tau_i^{-1}\tilde{\eta}_i\dot{\hat{\eta}}_i. \quad (57)$$

Based on the definition of W_σ and M, one can obtain that

$$\begin{aligned}&2\Gamma^T(M\otimes I_3)(W_\sigma\otimes I_3)\Gamma\\&=\Gamma^T\left[\left(W_\sigma^TM+MW_\sigma\right)\otimes I_3\right]\Gamma\\&\le\Gamma^T(Q_1\otimes I_3)\Gamma\end{aligned} \quad (58)$$

where

$$Q_1=\begin{bmatrix}-\epsilon\breve{\lambda} & 2\rho I_N-\epsilon\breve{\lambda} & (2+\mu)\rho I_N\\ 2\rho I_N-\epsilon\breve{\lambda} & 2\rho I_N-\epsilon\breve{\lambda} & (1+\mu)\rho I_N\\ (2+\mu)\rho I_N & (1+\mu)\rho I_N & -2\bar{\rho}\mu I_N\end{bmatrix}.$$

From (50) and (54), the following inequality can be achieved:

$$\begin{aligned}2\Gamma^T(M\otimes I_3)\mathbf{U}&=2\rho(\xi_h+\xi_v)^T[-c\Psi-(\mathbf{1}_N\otimes u_0)]\\&=-2\rho c(\xi_h+\xi_v)^T(\mathcal{H}_\sigma\otimes I_3)\mathbf{G}\\&\quad-2\rho(\xi_h+\xi_v)^T(\mathbf{1}_N\otimes u_0)\\&=-2\rho c\chi^TG-2\rho\chi^T\left(\mathcal{H}_\sigma^{-1}\mathbf{1}_N\otimes u_0\right)\\&\le-2\rho c\sum_{i=1}^{N}\frac{\|\chi_i\|^2\hat{\eta}_i^2}{\|\chi_i\|\hat{\eta}_i+\delta_i}+2\rho c\sum_{i=1}^{N}\eta_i\|\chi_i\|\end{aligned} \quad (59)$$

where η_i is the ith element of the vector $c^{-1}\lambda_{\min}^{-1}(\mathcal{H}_\sigma)\bar{u}\mathbf{1}_N$.

Since the nonlinear function $f(\cdot)$ satisfies the Lipschitz condition, it can be proved that

$$2\Gamma^T(M\otimes I_3)\Theta=2\bar{\rho}\zeta^T(t)\left[f(v)-f\left(\hat{v}\right)\right]\le2\varsigma_1\bar{\rho}\zeta^T\zeta \quad (60)$$

where $\varsigma_1=\sqrt{3}\max\{c_x/m,c_y/m,c_z/m\}$.

It also follows from (51) that:

$$\begin{aligned}\sum_{i=1}^{N}\varrho_i^{-1}(\epsilon_i-\epsilon_0)\dot{\epsilon}_i&=\sum_{i=1}^{N}(\epsilon_i-\epsilon_0)\chi_i^T\chi_i\\&\le\Gamma^T(Q_2\otimes I_3)\Gamma\end{aligned} \quad (61)$$

where

$$Q_2=\begin{bmatrix}(\epsilon-\epsilon_0I_N)\breve{\lambda} & (\epsilon-\epsilon_0I_N)\breve{\lambda} & 0\\(\epsilon-\epsilon_0I_N)\breve{\lambda} & (\epsilon-\epsilon_0I_N)\breve{\lambda} & 0\\0&0&0\end{bmatrix}.$$

On the other hand, from (52), one has

$$2\rho c\sum_{i=1}^{N}\tau_i^{-1}\tilde{\eta}_i\dot{\hat{\eta}}_i=2\rho c\sum_{i=1}^{N}\tilde{\eta}_i\|\chi_i\|-2\rho c\sum_{i=1}^{N}\tilde{\eta}_i\hat{\eta}_i\delta_i. \quad (62)$$

Considering the inequality $0\le(xy/[x+y])<y\quad\forall x\ge0,y>0$, and from (57)–(62), it follows that:

$$\dot{V}_h\le\Gamma^T\left(\tilde{Q}\otimes I_3\right)\Gamma+\varsigma_2\sum_{i=1}^{N}\delta_i \quad (63)$$

where $\varsigma_2=(2+[1/2]\eta_i^2)\rho c$ and $\tilde{Q}=\begin{bmatrix}\tilde{Q}_1&\tilde{Q}_2\\\tilde{Q}_2^T&\tilde{Q}_3\end{bmatrix}$, and

$$\tilde{Q}_1=\begin{bmatrix}-\epsilon_0I_N\breve{\lambda} & (2\rho-\epsilon_0\breve{\lambda})I_N\\(2\rho-\epsilon_0\breve{\lambda})I_N & (2\rho-\epsilon_0\breve{\lambda})I_N\end{bmatrix}$$

TABLE II
PARAMETER VALUES OF QUADROTOR UAV

Parameter	Value	Parameter	Value
m	2.5 kg	$\mathcal{I}_r$	1.39×10^{-4}kg·m^2
l	0.225 m	b	1.29×10^{-5}N·s^2/rad^2
g	9.8 m/s^2	d	1.74×10^{-7}N·m·s^2/rad^2
$\mathcal{I}_{x,y}$	0.0291 kg·m^2	$c_{x,y,z}$	0.01 N·s/m
$\mathcal{I}_z$	0.0467 kg·m^2	$c_{\phi,\theta,\varphi}$	0.01 N·s/rad

$$\tilde{Q}_2=\begin{bmatrix}(2+\mu)\rho I_N\\(1+\mu)\rho I_N\end{bmatrix},\quad\tilde{Q}_3=2\bar{\rho}(\varsigma_1-\mu)I_N.$$

Then, choose appropriate $\epsilon_0>2/\breve{\lambda}$ and $\mu>\varsigma_1$ such that $\tilde{Q}_1<0$ and $\tilde{Q}_3<0$. For sufficiently large $\bar{\rho}$, the inequality $\tilde{Q}_1-\tilde{Q}_2\tilde{Q}_3^{-1}\tilde{Q}_2^T<0$ holds. From Lemma 2, one can obtain that $\tilde{Q}<0$.

It follows from (63) that:

$$\begin{aligned}0\le V_h(t)&=V_h(t_0)+\int_{t_0}^{t}\dot{V}_h(s)ds\\&\le V_h(t_0)+\varsigma_2\sum_{i=1}^{N}\bar{\delta}_i\end{aligned} \quad (64)$$

which indicates that all states are globally bounded.

In addition, from (63), one can obtain that

$$\lim_{t\to+\infty}\int_{t_0}^{t}\lambda_{\min}\left(-\tilde{Q}\right)\|\Gamma(s)\|^2ds\le V_h(t_0)+\varsigma_2\sum_{i=1}^{N}\bar{\delta}_i. \quad (65)$$

Then, applying Barbalat's lemma, it can be proved that $\lim_{t\to+\infty}\|\Gamma(t)\|^2=0$. Therefore, Γ asymptotically converges to zero. The proof is completed. ■

Remark 10: If the distributed formation tracking controller $u_i=[u_{x_i},u_{y_i},u_{z_i}]^T$ is given by (48), and the estimated signal $\hat{\varphi}_0$ is treated as an extra desired signal, the desired roll angle $\phi_{i,0}$, the desired pitch angle $\theta_{i,0}$, and the control input U_{s_i} can be obtained according to (6) in the following way:

$$\begin{cases}\theta_{i,0}=\arctan\left(\frac{C_{\hat{\varphi}_0}u_{x_i}+S_{\hat{\varphi}_0}u_{y_i}}{u_{z_i}+g}\right)\\\phi_{i,0}=\arctan\left(\frac{\left(S_{\hat{\varphi}_0}u_{x_i}-C_{\hat{\varphi}_0}u_{y_i}\right)C_{\theta_{i,0}}}{u_{z_i}+g}\right)\\U_{s_i}=\sqrt{u_{x_i}^2+u_{y_i}^2+\left(u_{z_i}+g\right)^2}\end{cases} \quad (66)$$

where $\phi_{i,0}$ and $\theta_{i,0}$ are the desired roll and pitch angles of the ith UAV, respectively.

V. APPLICATION TO QUADROTOR UAV FORMATION FLYING

In this section, a swarm system consisting of five quadrotor UAVs is considered to verify the effectiveness of the proposed formation control scheme. The parameter values of each UAV are shown in Table II.

A. Comparison Experiment

In this section, the performance comparison between the distributed cascade robust control method [7] and the method proposed in this article is given. The communication topology of the quadrotor UAV swarm is shown in Fig. 4.

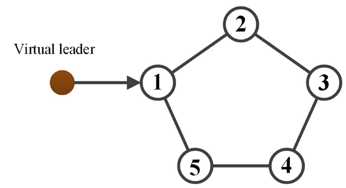

Fig. 4. Communication topology $\mathcal{G}$.

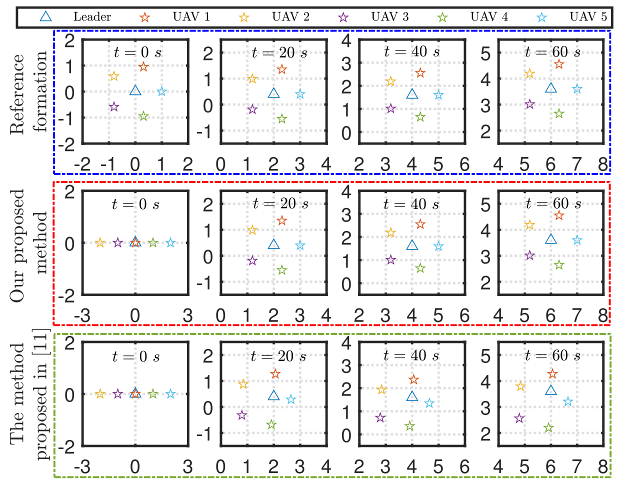

Fig. 5. Formation shapes in the X–Y plane.

A virtual leader is introduced to guide the quadrotor UAV swarm to fly along the desired trajectory. The system dynamics of the virtual leader are given as follows:

$$h_0(t) = [0.1t, 0.001t^2, 0.1t]^T, \quad \varphi_0 = t.$$

At the initial moment, five quadrotor UAVs are arranged in a line, and their initial positions are

$$h_1(0) = [0, 0, 0]^T, \quad h_2(0) = [-2, 0, 0]^T$$
$$h_3(0) = [-1, 0, 0]^T, \quad h_4(0) = [1, 0, 0]^T$$
$$h_5(0) = [2, 0, 0]^T.$$

In addition, during the flight of the quadrotor UAV swarm, all UAVs fly in the shape of a pentagon. The formation shape vectors are given by

$$F_{h_i}(t) = \left[\cos\left(\frac{2i\pi}{5}\right), \sin\left(\frac{2i\pi}{5}\right), 0\right]^T, \quad i = 1, \ldots, 5.$$

The proposed formation control scheme includes attitude-loop control and position-loop control. The controller parameters in the attitude loop are selected according to the design process in Section III. The parameters in the distributed attitude observer are selected as $\ell = 2.5$ and $\gamma = 0.1$. The parameters in the attitude controller are selected as $\lambda_{i,j} = 0.1$ ($j = 1, 2, 3$), $b_{i,1} = b_{i,2} = 4$, and $k_{m_{i,j}} = k_{n_{i,j}} = 0.5$. The controller parameters in the position loop are selected according to the design process in Section IV. The parameter in the velocity observer is selected as $\mu = 1.2$. The parameters in the formation tracking controller are selected as $c = 2$, $\varrho_i = 5$, $\tau_i = 2$, $\sigma_i(t) = 0.1e^{-t}$, $\epsilon_i(0) = 0$, and $\hat{\eta}_i(0) = 0$.

The reference formation shapes are shown in the first row of Fig. 5. The formation shapes obtained by our proposed method and the distributed cascade robust control method [7] are shown in the second and third rows of Fig. 5, respectively. The controller parameters of the method in [7] are appropriately selected as $\mathbf{K}^f = -2I_3$ and $\mathbf{K}^v = -2I_3$. As can be seen from Fig. 5, under the proposed formation control scheme, the quadrotor UAV swarm can form the desired formation and maintain the formation shape in motion. In contrast, the method in [7] cannot achieve the desired formation flight of the quadrotor UAV swarm within 60 s. Therefore, our proposed method can achieve better formation control performance than the method in [7].

In addition, under the proposed formation control scheme, the quadrotor UAVs' flight trajectories in the 3-D space are displayed in Fig. 6. It can be seen that five quadrotor UAVs reach the desired formation shape and track the virtual leader. The attitude angle response curves of the quadrotor UAVs are shown in Fig. 7. From Fig. 7, it is observed that the quadrotor UAV's yaw angle can track the desired yaw angle of the virtual leader. Fig. 8 shows the response curves of the formation tracking errors. It is noted that the formation tracking error of the quadrotor UAV finally converges to zero. That is, the formation tracking control of the quadrotor UAV swarm is achieved. The response curves of the adaptive parameters $\epsilon_i(t)$ are shown in Fig. 9. It can be seen that all adaptive parameters $\epsilon_i(t)$ eventually converge to some constants.

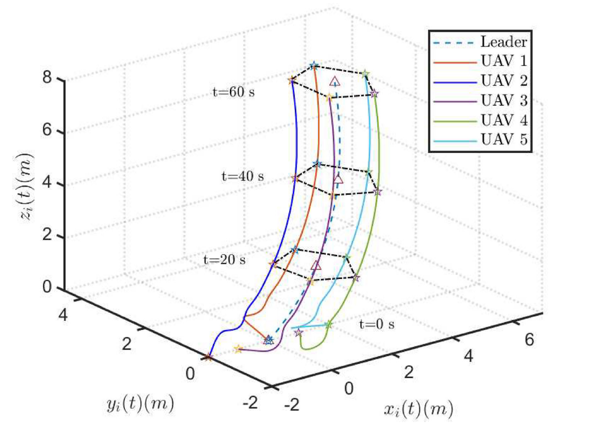

Fig. 6. UAVs' trajectories in the 3-D space.

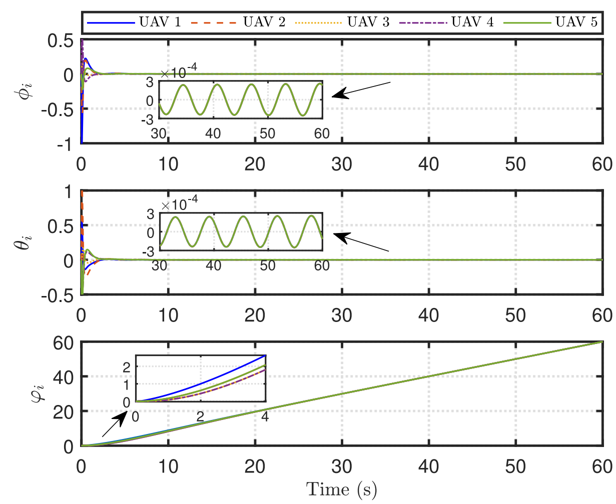

Fig. 7. Response curves of the attitude angles.

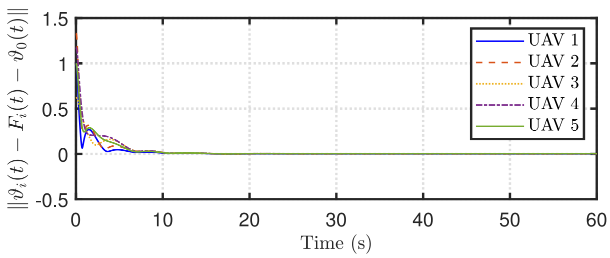

Fig. 8. Response curves of the formation tracking errors.

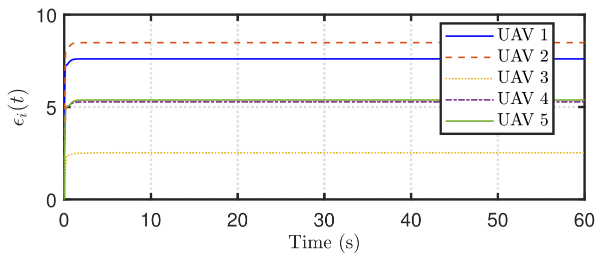

Fig. 9. Response curves of the adaptive parameters $\epsilon_i(t)$.

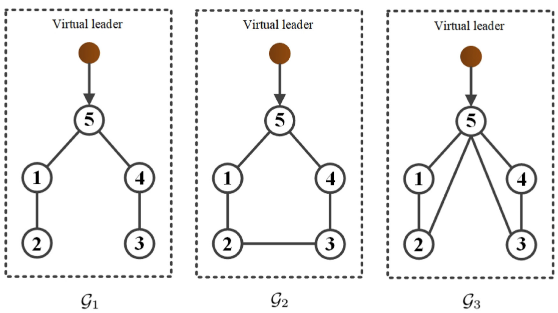

Fig. 10. Switching communication topologies $\mathcal{G}_s$ $(s = 1, 2, 3)$.

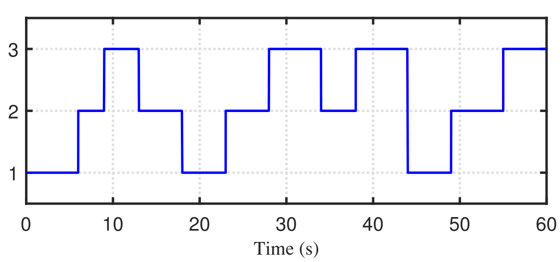

Fig. 11. Switching signal σ.

B. Switching Communication Topology Experiment

In this section, formation tracking control of a quadrotor UAV swarm with switching communication topologies is considered. The communication topology set $\mathbb{G} \triangleq \{\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3\}$, and all possible communication topologies are shown in Fig. 10. The random switching signal σ is shown in Fig. 11.

The system dynamics of the virtual leader are given as follows:

$$h_0(t) = [0.1t, 0.1t, 0.1t]^T, \quad \varphi_0 = t.$$

At the initial moment, five quadrotor UAVs are arranged in a line, and their initial positions are

$$h_1(0) = [0, 0, 0]^T, \quad h_2(0) = [-2, 0, 0]^T$$
$$h_3(0) = [-1, 0, 0]^T, \quad h_4(0) = [1, 0, 0]^T$$
$$h_5(0) = [2, 0, 0]^T.$$

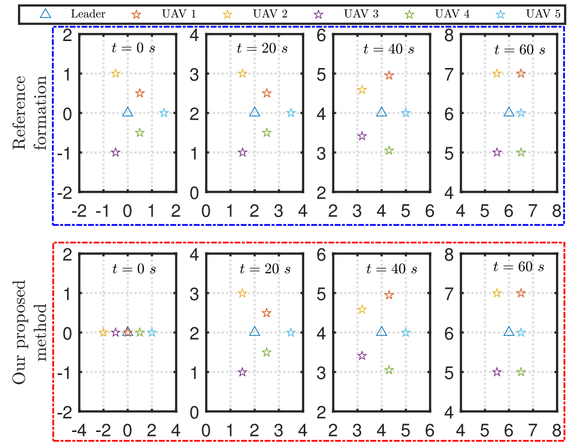

Fig. 12. Formation shapes in the X–Y plane.

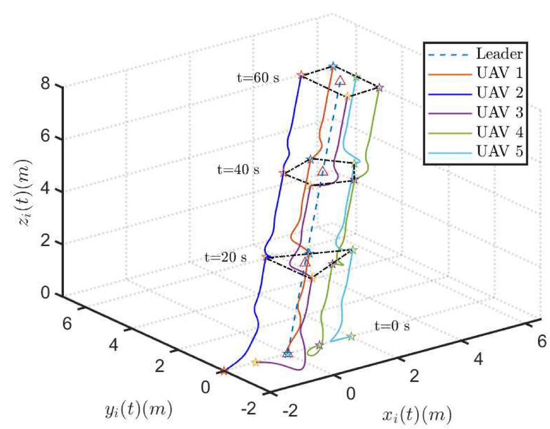

Fig. 13. UAVs' trajectories in the 3-D space.

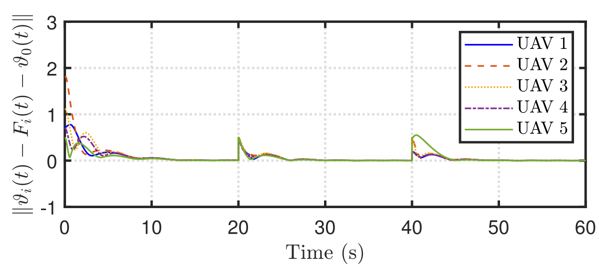

Fig. 14. Response curves of the formation tracking errors.

In addition, the time-varying formation shape vectors are given by

$t < 20s$:
$$F_{h_1}(t) = [0.5, 0.5, 0]^T, \ F_{h_2}(t) = [-0.5, 1, 0]^T$$
$$F_{h_3}(t) = [-0.5, -1, 0]^T, \ F_{h_4}(t) = [0.5, -0.5, 0]^T$$
$$F_{h_5}(t) = [1.5, 0, 0]^T$$
$20s \leq t < 40s$:
$$F_{h_i}(t) = \left[\cos(\tfrac{2i\pi}{5}), \sin(\tfrac{2i\pi}{5}), 0\right]^T, \quad i = 1, \dots, 5$$
$40s \leq t$
$$F_{h_1}(t) = [0.5, 1, 0]^T, \ F_{h_2}(t) = [-0.5, 1, 0]^T$$
$$F_{h_3}(t) = [-0.5, -1, 0]^T, \ F_{h_4}(t) = [0.5, -1, 0]^T$$
$$F_{h_5}(t) = [0.5, 0, 0]^T.$$

The formation controller parameters are the same as those in Section V-A. In addition, in order to verify the effectiveness of the proposed CADRC scheme, some time-varying disturbances are considered in the attitude subsystem of the quadrotor UAV, where $w_{m_{i,j}} = 0.5\cos(t)$ and $w_{n_{i,j}} = 0.5\sin(t)$ $(i = 1, \ldots, 5; j = 1, 2, 3)$. These time-varying disturbances satisfy Assumption 3.

The reference formation shapes are shown in the first row of Fig. 12. The formation shapes obtained by the proposed control scheme are shown in the second row of Fig. 12. As can be seen from Fig. 12, the proposed control method can achieve the time-varying formation control of the quadrotor UAV swarm. The quadrotor UAVs' flight trajectories in the 3-D space are displayed in Fig. 13. It can be seen that five quadrotor UAVs can achieve time-varying formation flight and track the virtual leader. The response curves of the formation tracking errors are shown in Fig. 14. It is observed that even when the reference formation changes, the formation tracking errors can still converge to zero in a relatively short time. In addition, as can be seen from Figs. 13 and 14, even if time-varying disturbances are introduced, the time-varying formation flight of the quadrotor UAV swarm can still be achieved. Therefore, the proposed control scheme is robust to time-varying disturbances.

VI. Conclusion

In this article, a distributed formation tracking control strategy that only relies on the position information of each UAV has been developed for a quadrotor UAV swarm system. A cascade ADRC method has been designed for the attitude subsystem to suppress the influence of unknown disturbances. With the proposed formation control approach, the formation tracking control of the quadrotor UAV swarm can be achieved, and all states of the swarm system are globally bounded. The proposed formation control approach has been verified by the performance analysis of the quadrotor UAV swarm system. The future work includes time-varying formation tracking control of heterogeneous quadrotor UAV swarms under more general communication topologies.

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